SINLGE USER BEng, MEng, MSc GROUP PROJECT RISK ASSESSMENT FORM - REPORT ONLY SIGNIFICANT HAZARDS

Unsafe working methods will lead to a reduction in your final project mark! ALL hardware work must be completed within the laboratory

Students are encouraged to come on site to perform their lab work but are advised that in some circumstances (Adriano, raspberry Pi and micro-controller boards which operate at <20V) equipment is allowed to be brought home. Students removing any other equipment from the lab needs to be authorised in writing by your supervisor - supervisors please confirm with HOD/safety team to confirm.

NAME- **Student ID Number- 201735577**	**LOCATION-** **Final year Laboratory**
SCHOOL/DEPARTMENT: Electrical Engineering & Electronics **Undergraduate year of study : 3**	**BUILDING: Electrical Engineering and Electronics, A-Block**
TITLE OF PROJECT: Advanced Fault-Tolerant Control for Attitude Stabilisation in CubeSats	
Description of Work: **Simulation and hardware testing of redundant stabilisation and control of lab-scale satellites using reaction wheels.**	
Select a category for this project: **Category 1 / 2 / 3**	**<u>Category 1 – Projects based on specialist equipment:</u>** Projects requiring equipment available in the electronics laboratories (such as power supplies, multimeters, oscilloscopes, etc.) or any other specialist equipment that requires specific health and safety considerations (such as drones, etc.) that students would not normally be allowed to take home.
	<u>Category 2 – Projects based on "home-friendly" equipment:</u> Projects requiring small pieces of equipment that do not require specific health and safety considerations and students can safely use at home (Raspberry Pi's, Arduinos and other similar low-voltage boards with double insulated power supplies).
	<u>Category 3 – Projects based on software only:</u> Projects fully based on software that can be completed using only a computer, without requiring any other equipment.

If students are in an observation capacity only when experiment is being performed

- please state this on form as well as risk in being observers - i.e. possible distracting experimentalist,
- State risk if they could be injured in this respect and how. Significant risks only should be stated.
- Class of any laser is required

State voltage & current values of all power sources being used. Any power supplies that have the ability to generate current and voltages > 10mA AND >20V respectively can be regarded as potentially extremely hazardous:

Voltage Current

HAZARDS (Location, equipment and substances, activities)	WHO CAN BE HARMED?	CURRENT CONTROLS	Likelihood (L) × Consequence (C) = RISK SCORE (R)		
			L	C	R
Soldering Iron heat (400° C) and solder fumes	**User, other students**	**- Ensure use of fume extractor to reduce risk of occupational asthma. - Ensure iron is handled carefully and allowed to cool between uses. - Use lead free solder to reduce respiratory damage - Switch off when not in use. - Do not leave soldering iron unintended while in use or when cooling down.**	2	1	2
Motors / Robotics causing finger traps, snatches or snags	**User, nearby observers**	**- Ensure loose hair/clothing is tied back or removed. - Keep adequate distance and do not touch device when powered on. - Enclose moving parts to reduce the risk of entanglement or ejection of loose material. - Ensure device is built to a high quality, and inspected for mechanical faults before use. - Ensure design prevents fingers from being able to contact with moving parts when enclosed. - Shield between motors**	1	2	2
Heat / Fire from motor/electronics	**User, nearby staff/students**	**- Ensure that motors and motor electronics are provided adequate cooling/ open contact with air. - Check connections several times for shorts or wrong connections to reduce likelihood of thermal runaway. - Limit current to motor driver to limit heating in both motor and electronics.**	2	2	4
DSE and Programming strains	**User**	**- Ensure frequent breaks are taken to reduce the likelihood of repetitive strain injury, carpal tunnel syndrome or fatigue.**	1	1	1
Noise (disturbance) and Vibration (disassembly)	**User, nearby staff/students**	**- Inspect hardware before testing to ensure there are no loose components. When testing hardware, run the motors well below their maximum throttle to reduce noise. - Use dampening material between plates to reduce the effect of vibrations and noise - Use an enclosure or box to dampen noise (with foam dampening for example) - Use spring washers or adhesive to lock screws that are made loose during operation - If motors are still noticeably loud, run for short periods to reduce disturbance too others - If needed, relocate to an appropriate environment and use hearing PPE (Very unlikely to be needed)**	2	2	4
Thermal runaway due to over-charge or discharge of battery	**User, nearby staff/students**	**- Store batteries in dedicated fire proof bag - Charge and discharge batteries at the recommended rate outlined by the manufacturer, using a Lipo balance charger rated for the same cell count.**	1	3	3
Over current leading to fire	**User, nearby staff/students**	**- Ensure connectors and wiring are rated for the maximum expected current**	1	2	2
Fire due to physical damage to battery	**User, nearby staff/students**	**- House the battery in rigid housing so that it cannot be accidentally dropped or punctured while in use**	1	3	3

Fire due to chemical damage of battery (Undervoltage)	**User, nearby staff/students**	- **Ensure the cell voltage does not sink below the cells rated minimum voltage. Enforce b powering off electronics**	1	2	2
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		-			
		-			
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- **For work using only Raspberry Pi and/or Arduino boards or other hardware connected via USB cable the main hazards are Display Screen Equipment (DSE) related, e.g. Repetitive Strain Injury, Carpal Tunnel Syndrome. L=1, C=1, R=1**

Training table - All boxes must be ticked in the following section to indicate either YES or NO.			
	N0	**YES**	**If you have ticked YES please follow the hyperlinks in the attached document, complete and return supplementary paperwork and/or implement and adhere to the guidance given.**
Use of tenon saw/hacksaw	✓		**Read Safe Operating** Procedure and other documentation on hand tools
Will work require the lifting of weights (>15kg)	✓		Manual Handling
Laser – If yes please input class of laser. Laser documents and hazard should be described on page 2 if laser is **NOT** class 1	✓		Please read all documents in the following link README : Laser: information and registration Guidance on the Safe Use of Lasers in Education & Research
Use of drones	✓		Prior to the purchase of any drone equipment, please consult the safety team – even if the devices purchased are not intended to be flown. Please refer to drones
Use gas cylinders or compressed gas?	✓		Gas Cylinder safety : Email local safety team to verify if training is required
Use hazardous Chemicals only? If stated on the form, description of hazard is required.	✓		COSHH - Use on-line EEE COSHH system to create COSHH risk assessment. Email local safety team to verify if training is required
Use voltages over 30V DC/AC If hazard has been previously described this	✓		Electrical Safety/Electricity – Includes reading the Dept. of EEE dangers of electricity document
Use Power tools or rotating motors and machines	✓		SCR15-4 PUWER
Use Cryogenic Liquids/gases	✓		Cryogenic liquids and solids – Email local safety team to verify if training is required
Use Vacuum Systems and pressurised vessels	✓		Pressure systems : Email local safety team to verify if training is required
Use Radiation (UV, x-rays, microwaves)	✓		UV radiation (including links to local rules & safety advisor website)

LEVEL of Supervision?	A = Work May not be started without direct supervision
	B = Work may not start without Supervisor advice or approval
	C = No specific extra supervision requirements

Other relevant specific assessments (Local rules, Ethic approval forms)-

Disclaimer

- *The University of Liverpool ensures as far as is reasonably practical the health and safety of its staff and students.*
- *All equipment used by the students for their project must be safety tested and approved by the laboratory technicians before use. This includes but is not limited to, soldering irons, oscilloscopes, power supplies, probes and multimeters.*
- *Students **MUST NOT** undertake hazardous experimental/development work associated with their project outside of their designated laboratory space.*
- *ALL equipment that is used in the laboratory space & project MUST be purchased through the departments purchasing procedures.*
- *No equipment to be plugged into the mains supply unless circuit has been approved by technician or supervisor.*
- *Failure to abide by these conditions can result in the project receiving 0%.*
- ***Submission of this form implies acknowledgement by all the students named below.***

I can confirm that Hazards identified and precautions specified are appropriate for the task :-

Acknowledgement by Student Name Matteo Venuti Signature Date 04/10/25

Academic supervisor Name Barry Smith Signature Date 04/10/25

Common reasons for previously rejection of the form

- Project category was not stated on the assessment.
- Contradiction of hazards listed on page 2 compared those identified in training table. Users inserted description of hazards such as chemicals & live working but failed to insert yes in hazard table. Only hazardous chemicals should be described. Only significant hazards observed in experimental process should be described.
- Missing supervisor signature – risk assessment is invalid & students cannot enter the laboratory area
- Additional hazards noted in training table that are not described in hazard section. Lasers were described in training table required but hazard was not described in main assessment. Laser users should refer to risk assessment template document to identify how these should be described.

GUIDANCE TO COMPLETE THIS RISK ASSESSMENT FORM (LIKELIHOOD / CONSEQUENCE / RISK SCORE)

Likelihood			Consequence			Risk score	ACTION TO BE TAKEN
1	Very unlikely		1	Insignificant – no injury		1-2 NO ACTION	No action required but ensure controls are maintained and reviewed.
2	Unlikely		2	Minor – minor injuries needing first aid		3-9 MONITOR	Look to improve at next review of if there is a significant change
3	Fairly likely		3	Moderate – up to seven days absence		8-12 ACTION	Reduce risk if possible, within specified timescale
4	Likely		4	Major – more than seven days absence; major injury		15-25 STOP	Stop activity and immediate action
5	Very likely		5	Catastrophic – death; multiple serious injury			

- For work using only Raspberry Pi and/or Arduino boards (i.e. no other hardware connected using additional power supplies) the only hazards are Display Screen Equipment (DSE) related, e.g. Repetitive Strain Injury, Carpal Tunnel Syndrome. L=1, C=1, R=1

Increasing Consequence ↑					
5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5
	1	2	3	4	5

Increasing Likelihood →

15-25	**Stop** Stop activity & immediate action. Must seek advice
8-12	**Action** Improve within specified timescale
3-6	**Monitor** Look to improve at next review or if there is a significant change
1-2	**No Action** No action required but ensure controls are maintained & reviewed